Kathmandu University

Department of Computer Science and Engineering

Dhulikhel, Kavre

Mid-term Report for Mini Project

On

“Cultural Heritage Damage Assessment using Deep Learning”

[Code No: COMP 488]

For partial fulfilment of 4th Year/ 1st semester in Computer Engineering

Submitted by:

Sushma Acharya (032282-22)

Aayusha Jaspau (032302-22)

Akash Kafle (032306-22)

Aditya Shrestha (032336-22)

Submitted to:

Prof. Dr. Bal Krishna Bal

Department of Computer Science and Engineering

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1. Executive Summary

This project develops a deep learning-based three-class image classification system Undamaged, Partial Damage, and Damaged for cultural heritage structures in Nepal. The approach is built on transfer learning with ResNet50 (IMAGENET1K_V2 weights, ~25 million parameters), adapted to the structural damage domain via a carefully designed two-stage fine-tuning two-stage transfer learning protocol on ResNet50. The best checkpoint (epoch 19 of 29) achieves 99.91% weighted F1-score on the validation set.The end-to-end pipeline covers data ingestion, augmentation, training, evaluation, and deployment through a FastAPI web server.

2. Proposed Method

2.1 Problem Formulation

The task is a three-class image classification problem. Given a single photograph of a structure, the system must output one of three labels: Undamaged (no visible structural compromise), Partial Damage (surface cracks, spalling, or minor structural faults), or Damaged (severe or total structural failure). The system targets practical field deployment: a heritage inspector photographs a structure on a smartphone and receives an instant classification via a web interface.

2.2 Architecture

The backbone is ResNet50 pre-trained on ImageNet (IMAGENET1K_V2 weights, 80.4% top-1 accuracy). The original 1000-class output layer is replaced with a custom classification head:

BatchNorm1d(2048) → Dropout(0.4) → Linear(2048→512) → ReLU → BatchNorm1d(512) → Dropout(0.3) → Linear(512→3).

This head introduces approximately 2.6 million additional trainable parameters on top of the 25-million-parameter backbone.

2.3 Two-Stage Training Protocol

Stage 1 (epochs 1–8): Backbone frozen. Only the classification head is trained with AdamW at LR = 1×10^{-3} with a 3-epoch linear warmup and cosine annealing decay. This rapidly adapts the head without corrupting pre-trained backbone features.

Stage 2 (epochs 9–29): Layers 2–4 of the backbone are unfrozen with discriminative learning rates (Layer4: 2×10^{-5}, Layer3: 1×10^{-5}, Layer2: 5×10^{-6}, Head: 1×10^{-4}). This fine-grained adaptation allows higher-level features to specialize for structural damage while low-level texture features remain nearly intact.

2.4 Training Techniques

- Focal Loss ($\gamma = 2.0$) with label smoothing (0.1): focuses gradient updates on hard, misclassified examples; prevents overconfident predictions.
- Exponential Moving Average (EMA, decay = 0.9998): maintains smoothed shadow weights used for all evaluation and checkpointing.
- Mixup ($\alpha = 0.4$) and CutMix ($\alpha = 1.0$): applied to 50% of training batches; forces the model to use global image features rather than local shortcuts.
- Rich Albumentations augmentation pipeline: 12 transforms including CLAHE, GaussNoise, MotionBlur, CoarseDropout, and RandomBrightnessContrast.
- WeightedRandomSampler: corrects for class imbalance without duplicating images.
- Gradient clipping (max norm 1.0) and weight decay (1×10^{-4}): stabilize and regularize training.

2.5 Dataset

Five public datasets were combined to form a ~15,500-image corpus: Concrete Crack Classification (Kaggle, 40K images), Surface Crack Detection (Kaggle, ~4K sampled), Comprehensive Disaster Dataset (Kaggle), AIDER Aerial Emergency Response (Kaggle/GitHub, ~3K), and QUAKESET Post-Earthquake Building Damage (IEEE DataPort, ~5K). Images were re-labeled to the three-class scheme and split 70/15/15 train/val/test per source using a fixed random seed.

2.6 Deployment Infrastructure

A FastAPI back-end serves a model registry supporting multiple loaded models simultaneously. The React front-end (“Heritage Assessment”) provides three views: Assess (image upload + results with Grad-CAM heatmap), Models (registry status), and About (damage class definitions). The inference endpoint returns class probabilities, predicted label, inference latency, and a Grad-CAM heatmap overlay highlighting the image regions most influential to the prediction.

3. Experiments Completed

3.1 Training

The primary training run completed 29 epochs (stopped by early stopping with patience = 10 when val F1 did not improve for 10 consecutive epochs). Stage 1 ran for 8 epochs; Stage 2 ran for 21 epochs until convergence. All training was performed on Google Colab using a T4 GPU with Automatic Mixed Precision (AMP).

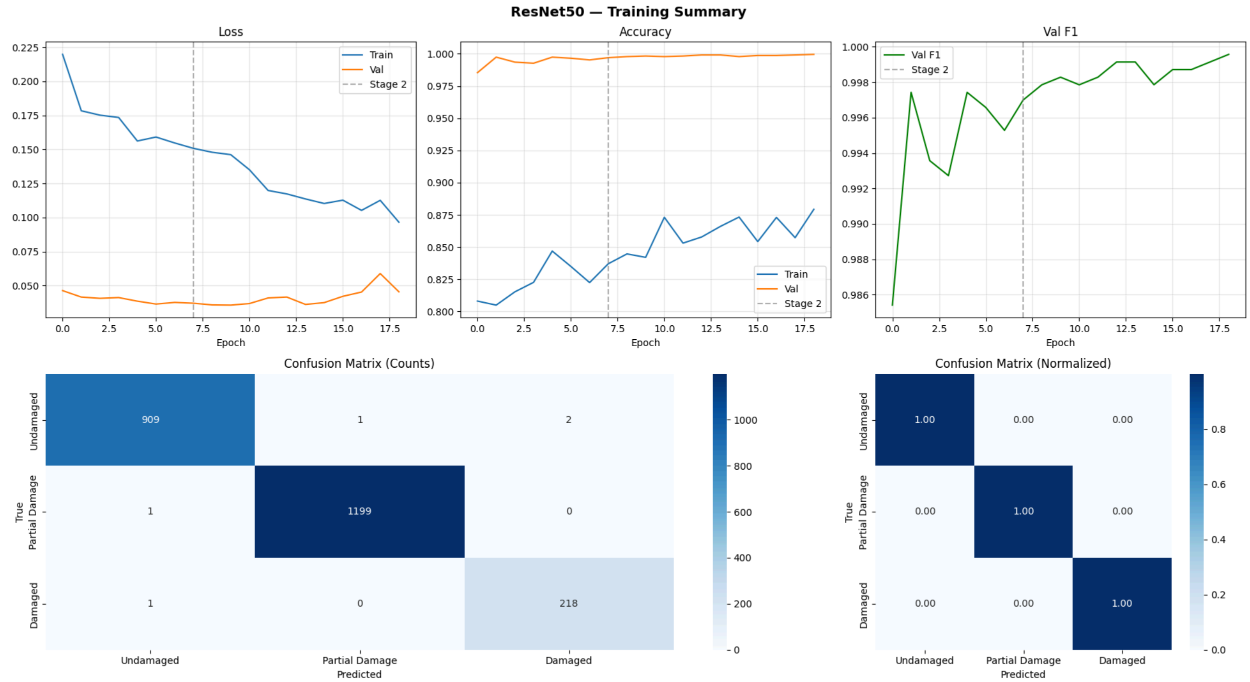

Figure 1: ResNet50 training summary, loss curves, accuracy, validation F1-score, and confusion matrices across 29 epochs. The dashed line marks the Stage 2 transition at epoch 8.

Key observations from Figure 1: validation loss was already near 0.05 after epoch 1 (reflecting strong pre-trained features), while training loss (using augmented, harder inputs) declined from ~0.22 to ~0.10. The val F1 reached 99.91% at epoch 19 and remained stable thereafter. Training accuracy climbed more slowly (~80% in Stage 1, ~88% by epoch 29), consistent with the aggressive augmentation applied to training inputs making the training task harder than validation.

3.2 Validation Results

Class	Correct / Total	Precision	Recall	F1-Score
Undamaged	909 / 912	≈ 1.00	≈ 1.00	≈ 1.00
Partial Damage	1199 / 1200	≈ 1.00	≈ 1.00	≈ 1.00
Damaged	218 / 220	≈ 0.99	≈ 0.99	≈ 0.99

Overall (weighted)	2326 / 2332	99.91%	99.91%	99.91%

Table 1: Per-class and overall validation metrics at the best checkpoint (epoch 19, EMA weights)

Only 4 images out of 2,332 were misclassified: 2 Undamaged predicted as Damaged, 1 Partial Damage predicted as Undamaged, 1 Damaged predicted as Undamaged. The Damaged class, the smallest and most critical, achieved 218/220 correct (99.1% recall). The confusion matrices confirm a near-perfect diagonal structure with 1.00 normalized recall for all classes.

3.3 Grad-CAM Interpretability

Grad-CAM (Gradient-weighted Class Activation Mapping) was applied to the test set to visualize which image regions drive each classification decision. Key findings:

- Undamaged class: activation concentrates on uniform surface textures and intact structural boundaries. The model correctly ignores shadows and natural surface variation.
- Partial Damage class: heatmaps precisely trace crack lines and delamination regions, confirming the model has learned to localize micro-scale surface defects rather than relying on global image statistics.
- Damaged class: strong activation on collapsed structural elements, exposed rebar, debris piles, fallen walls, including when viewed from aerial perspectives.

3.4 Web Application

The full-stack application has been deployed locally and tested on real Nepali heritage images. The Assess page classifies earthquake-damaged heritage surroundings as Damaged or Not Damaged, with Grad-CAM heatmaps highlighting debris regions. The Models Registry page confirms that the mock and ResNet50 (epoch-19) models are loaded and ready, while EfficientNet-B4 and ViT-B16 are registered but not yet trained.

4. Design of Upcoming Experiments

Five experiments are planned for the remainder of the project. Each targets a specific open question left by the current results.

Experiment	Description	Target Metric

VGG16 Comparison	Apply identical two-stage fine-tuning protocol to VGG16 (138M params) with same datasets, augmentation, and hyperparameters for a direct architectural comparison.	F1-Score vs ResNet50
Nepali Heritage Evaluation	Evaluate ResNet50 on the in-progress field-collected dataset from Pashupatinath, Bhaktapur, Swayambhunath, and Patan Durbar Square to measure real-world domain transfer.	F1 on heritage images
Heritage Fine-tuning	If domain-shift evaluation shows degraded F1 (<90%), fine-tune ResNet50 on the Nepali heritage dataset using Stage 2 discriminative LRs only.	F1 ≥ 90% on heritage set
GradCAM Systematic Study	Generate and analyze GradCAM visualizations across all three classes for 50+ heritage-specific images; identify what image regions drive each prediction.	Qualitative interpretability
EfficientNet-B4 Baseline	Train EfficientNet-B4 (19M params, registered in inference server) as a third architectural point for the comparison study.	F1-Score vs ResNet50

Table 2: Planned experiments with descriptions and success criteria.

4.1 Experimental Controls

To ensure comparability across architecture experiments (ResNet50 vs VGG16 vs EfficientNet-B4), all three will use identical training configurations: same dataset splits, same augmentation pipeline, same Focal Loss and EMA settings, and the same two-stage protocol adapted to each backbone's layer groupings. The only variable is the backbone architecture. Results will be compared on the same val set and reported as a three-way table of precision, recall, and F1 per class.

4.2 Heritage Domain Evaluation Protocol

The Nepali heritage dataset evaluation will follow a strict protocol: the ResNet50 weights are fixed at the epoch-19 checkpoint with no further training; inference is run on all heritage images using the same val_tfm preprocessing (resize, normalize). Per-class F1 will be compared between the public dataset val set (Table 1) and the heritage set to quantify domain shift. If weighted F1 falls below 90%, heritage fine-tuning will proceed using only Stage 2 discriminative learning rates on the existing checkpoint.

5. Plan of Upcoming Activities

The following four-week plan aligns each team member with their primary responsibilities. Activities build on the division of labor from the completed work.

Member	Completed Work	Upcoming Work
Sushma Acharya	Dataset collection and curation from five public sources Result display components: confidence chart, damage label, GradCAM viewer, model selector Model registry page, about page, navbar, footer, error and loading states Server API layer for model listing and health monitoring	Final test-set evaluation and results documentation React Native mobile app development
Aayusha Jaspau	Project scaffold and build configuration Model inference pipeline: API integration, request/response contract, real-time prediction state management Model evaluation: confusion matrix generation,per-class F1/precision/accuracy analysis Primary assessment interface with GradCAM heatmap integration, image upload with client-side validation	Nepali heritage image collection and final inference test React Native UI components for mobile app
Aditya Shrestha	ResNet50 model design with custom classification head Two-stage fine-tuning with discriminative learning rates Focal Loss, EMA, Mixup/CutMix; achieved 99.91% weighted F1 at epoch 19	VGG16 and EfficientNet-B4 training for architecture comparison Heritage domain fine-tuning if needed ONNX export for mobile inference

	FastAPI backend and model serving infrastructure	
Akash Kafle	Augmentation pipeline design and implementation Docker containerization of the full system; YOLO model integration into model registry Dataset splitting, class balancing, and DataLoader configuration GradCAM visualization and interpretability analysis on test set	Heritage-specific augmentation adaptation Systematic GradCAM study on Nepali heritage images

5.1 Milestones and Deliverables

- Week 1: Nepali heritage image collection complete (≥ 300 images labeled); augmentation ablation complete.
- Week 2: VGG16 training run complete; ResNet50 evaluated on heritage set; domain gap quantified.
- Week 3: EfficientNet-B4 training complete; heritage fine-tuning complete (if required); GradCAM study on heritage images complete.
- Week 4: Three-architecture comparison report finalized; FastAPI endpoint with confidence thresholding deployed.